• **DRIVEABILITY—BI-FUEL—HESITATION OR IDLE SURGE—HARD START, NO START OR START THEN STALL—WITH FUEL SELECTOR SWITCH LAMP ON IN B-FUEL MODE—VEHICLES EQUIPPED WITH 5.4L/6.8L BI-FUEL ENGINE ONLY** • **DRIVEABLITY—BI-FUEL—HARD START, NO START OR START THEN STALL, HESITATION OR IDLE SURGE—WITH FUEL SELECTOR SWITCH LAMP ON IN BI-FUEL MODE—VEHICLES EQUIPPED WITH 5.4L/6.8L BI-FUEL ENGINE ONLY**	**Article No. 01-6-5**

FORD: 2000 F-150, SUPER DUTY F SERIES

This TSB applies to 2000 F-150 LPG bi-fuel vehicles with compuvalve serial numbers between LG001054 and LG003379R and between LH001002 and LH002632R, and 2000 F-350/450/550 LPG bi-fuel vehicles with compuvalve serial numbers between LE000170 and LE001326R.

ISSUE

Some vehicles when driving in the alternative fuel mode may exhibit a hard start, no start or start then immediate stall, hesitation or surge at idle, with fuel selector switch lamp on in bi-fuel mode. This may be caused by improper sealing of the compuvalve fuel injectors.

ACTION

Verify condition, inspect and if necessary, replace compuvalve. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

NOTE

BEFORE BEGINNING THE SERVICE PROCEDURES, PERFORM THE FOLLOWING

PRELIMINARY CHECKS:

- Check for vacuum leaks including loose fittings, worn, chafed or crimped hoses and repair as required.
- Check recent work/repairs performed to ensure that the fuel induction system has not been disrupted and repair as required.
- Check the serial number of the compuvalve by reading the label on the side or top of the unit. If it does not fall within the serial number range stated, do not use this TSB, follow normal diagnostic procedures.
- Use New Generation Star (NGS) Tester to check hours of use in the alternative fuel mode (PID ALTFRUN). If ALTFRUN is greater than 100 hours, do not use this TSB, follow normal diagnostic procedures.
- Check for Diagnostic Trouble Codes (DTCs) and repair as required.

Refer to the following symptom descriptions to aid in diagnosis:

NO START OR STARTS THEN IMMEDIATELY STALLS (WITH FUEL SELECTOR SWITCH LAMP ON)

1. If the vehicle is a no start, attempt to start with throttle assistance by opening throttle to 1/4 open.
   a. If the vehicle cannot be started with throttle assistance, follow normal diagnostic procedures.

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b. If the vehicle can be started with throttle assistance, then replace compuvalve. (Refer to the appropriate Workshop Manual, Section 303-04E for compuvalve removal and installation procedures).

c. If the vehicle starts, then immediately stalls, replace compuvalve. (Refer to the appropriate Workshop Manual, Section 303-04E for compuvalve removal and installation procedures).

HARD START WITH SURGING IDLE - MAY ALSO STALL ON ACCELERATION (WITH FUEL SELECTOR SWITCH LAMP ON)

1. Start vehicle and use NGS Tester to access the high flow injector PIDs (PIDs VLV#1, VLV#2).

2. Monitor the injector states. VLV#1 should be OPEN, VLV#2 should be CLOSED. If the idle is surging, it will rapidly switch between VLV#1 OPEN, VLV#2 CLOSED and VLV#1 CLOSED, VLV#2 OPEN. A vehicle idling normally will stay steady on VLV#1 OPEN.

NOTE
UNDER CERTAIN CONDITIONS, THE VEHICLE MAY IDLE ON THE LOW FLOW INJECTORS ONLY (ALL VLV NUMBERS WILL READ CLOSED). IN ORDER TO FORCE VLV#1 OPEN, IT MAY BE NECESSARY TO INCREASE ENGINE LOAD BY TURNING ON ALL AUXILIARY EQUIPMENT (A/C, FAN, LIGHTS).

3. Increase rpm with transmission in Park or Neutral until VLV#2 reads OPEN.

   a. If the vehicle idles rough when VLV#1 reads OPEN and engine operation smoothes out when VLV#2 changes from closed to OPEN, then replace compuvalve. (Refer to the appropriate Workshop Manual, Section 303-04E for compuvalve removal and installation procedures).

   b. If the vehicle idles rough when VLV#1 reads OPEN and engine operation remains rough when VLV#2 changes from closed to OPEN, then follow normal diagnostic procedures.

SEVERE HESITATION ON ACCELERATION OR DECELERATION - MAY ALSO STALL (WITH FUEL SELECTOR SWITCH LAMP ON)

1. Start vehicle and use NGS Tester to access rpm and the high flow injector PIDs (PIDs RPM, VLV#1, VLV#2, VLV#3, VLV#4 and VLV#5).

NOTE
IT IS NOT NECESSARY TO MONITOR VLV#6.

2. Increase rpm until the injectors change from VLV#1 OPEN, VLV#2 CLOSED to VLV#1 CLOSED, VLV#2 OPEN. If a decrease of 100 rpm or more is indicated during the change from VLV#1 OPEN to VLV#2 OPEN, then replace compuvalve. (Refer to the appropriate Workshop Manual, Section 303-04E for compuvalve removal and installation procedures).

   a. If a decrease of 100 rpm or more is not indicated during the change from VLV#1 OPEN to VLV#2 OPEN, then continue to increase rpm to open each of the injectors in sequence.

NOTE
INJECTORS VLV#4 AND VLV#5 WILL REQUIRE A ROAD TEST TO ACHIEVE THE RPM REQUIRED TO OPEN THESE VALVES.

CAUTION
TWO TECHNICIANS ARE REQUIRED DURING THE ROAD TEST. DO NOT ATTEMPT TO DRIVE AND READ THE NGS TESTER ALONE.

   b. If a decrease of 100 rpm or more is indicated during any injector status change from CLOSED to OPEN, then replace compuvalve.

   c. If a decrease of 100 rpm or more is not indicated during any injector status change from CLOSED to OPEN, then follow normal diagnostic procedures.

PART NUMBER	PART NAME
YL3Z-9F954-TARM	Compuvalve (2000 F-150 LPG Bi-Fuel, 3.55)
YL3Z-9F954-UARM	Compuvalve (2000 F-150 LPG Bi-Fuel, 3.73)
YC3Z-9F954-ABRM	Compuvalve (2000 F-350/450/550 LPG Bi-Fuel)

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 404000, 602300, 603300, 607400, 607500, 607600, 607700, 608000